CHUKWUEMEKA OSMUND AZODO

ROTATING EQUIPMENT / MACHINERY RELIABILITY ENGINEER | OIL, GAS & LNG

B.Eng Mechanical Engineering | ASQ Certified Reliability Engineer (CRE) | COREN R.Engr | ASME, SPE, NSE, ASQ
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BOSIET with CA-EBS (valid to 2029) | Offshore Safety Permit (OSP) | **Available for immediate international mobilisation**

PROFESSIONAL PROFILE

Mechanical engineer with **10+ years' experience** in rotating equipment engineering across offshore gas processing, NGL trains, associated gas gathering and gas-fired power generation — currently the **Machinery Engineer for the Oso Gas Hub**, the largest offshore gas processing facility in the Seplat/JV portfolio. Day-to-day accountability for the **health assessment surveillance of critical rotating equipment** — HP/LP centrifugal gas compressors (5,500+ psi), turbo-expanders, industrial gas turbines, cryogenic and API 610 pumps, diesel engines and lubrication/seal systems — using vibration and lube-oil analytics, performance/compressor maps and digital platforms (SolarInsight, Seeq, PI XHQ). Works shoulder-to-shoulder with operations, maintenance, process, instrumentation and inspection teams to **diagnose machinery problems, optimise intervention scope and duration, and recommend engineered resolutions**. Leads and contributes to **RCFA, risk assessments and technical investigations**; reviews MOCs and project documentation for reliability and operability; and supports **FAT/SAT, commissioning and start-up** of new machinery. Established liaison with OEMs and vendors (Solar Turbines, GE, Siemens and Honeywell CCC). Strong working knowledge of **API 610, API 614, API 617, ISO 10816/20816 and ASME PTC-22**.

CORE TECHNICAL COMPETENCIES

Rotating Equipment	Centrifugal gas compressors (HP/LP, recompressors) • Turbo-expander and booster-compressor trains • Industrial gas turbines • Steam turbines • API 610 centrifugal and cryogenic pumps • Diesel engine drivers • Screw and reciprocating air compressors • Dry-gas and mechanical seals, journal/thrust bearings, couplings, lube-oil and seal-gas systems **(OEMs and models listed on page 3)**
Health Assessment & Surveillance	Vibration analysis (ISO 10816/20816, Cat-I) • Lube-oil sampling, elemental & particle-count analysis • Infrared thermography • Videoscope/borescope inspection • Performance monitoring via compressor maps, turbine heat rate and pump curves • Deviation flagging, trending and health reporting to stakeholders
Digital Tools & Programs	SolarInsight • Seeq • PI / PI XHQ historian & dashboards • Honeywell CCC anti-surge and turbomachinery control • SAP PM (notifications, work orders, strategy) • CMMS/EAM data quality • IIoT sensor-to-cloud (Modbus, OPC-UA) • Python/analytics and AI-assisted failure prediction
Reliability & Problem Solving	Root Cause Failure Analysis (RCFA) • RCA (5-Why, Fishbone, Fault Tree) • FMEA / RCM • Bad-actor and MTBF analysis • Criticality ranking • Life-cycle cost • Reliability improvement and defect-elimination programmes
Projects, Overhauls & Commissioning	Major overhaul planning and execution — gas and steam turbines, compressors, pumps, engines • Technical review of vendor/engineering documents, data sheets and deviations • Machinery FAT/SAT witnessing • Pre-commissioning, commissioning and start-up support • PSSR development and execution • Brownfield modification and capacity-expansion support (120 → 240 MMSCFD)
Operations Interface & HSE	Operations/maintenance decision support • Shutdown & turnaround scope optimisation • Risk assessment, HAZID/JSA participation • Permit to Work, LOTO, SIMOPS • OEM and vendor coordination • Multi-discipline technical support and mentoring
Standards	API 610 • API 614 • API 617 • ASME PTC-22 • ISO 10816 / 20816

PROFESSIONAL EXPERIENCE

Machinery Engineer Specialist II — Offshore Asset Engineering — May 2025 – Present

*Amaiden Energy Nigeria Ltd — seconded to **Seplat Energy Producing Nigeria Unlimited** (JV offshore assets: Oso Gas Hub, BRT, UBIT)*

Provides rotating equipment and reliability engineering support across three offshore assets; deployed February 2026 as the dedicated **Machinery Engineer for the Oso Gas Hub** (120 MMSCFD sales gas + NGL export, expanding to 240 MMSCFD), covering top-tier machinery (HP/LP compressors, recompressors, turbo-expanders, cryogenic pumps), utilities (gas turbine generators, firewater pumps, air compressors) and the NGL fractionation train (de-ethaniser, de-propaniser, de-butaniser).

Equipment Health Surveillance & Technical Support

- Run periodic **health assessment surveillance of critical rotating equipment** — review vibration spectra and trends, lube-oil analysis results, compressor performance maps, turbine parameters and pump deliverability curves — and issue analyses and recommendations to operations, maintenance and asset management for every deviation.
- Exploit digital solutions (**SolarInsight, Seeq, PI XHQ**) to trend machinery condition and process parameters, converting raw data into early-warning calls that prevent unplanned trips and deferment.
- Act as the discipline reference point for other engineering and non-engineering functions — process, instrumentation/controls, inspection, operations, planning, procurement and finance — providing rotating-equipment

technical input, spares criticality advice and specification support.

- Provide interim **Asset Engineer coverage**, extending beyond machinery into production optimisation, process stabilisation and non-machinery plant systems — ensuring uninterrupted technical support across the facility.

Machinery Problem Resolution, RCFA & Investigations

- **LP gas compressor (130 MMSCFD, 1,440+ psi) severe vibration after bundle change-out:** led the multi-discipline RCA with OEM and in-house teams — mapped contamination paths, ran sectioned borescope inspections of cooler bundles, identified debris-induced rotor unbalance, then directed process piping cleaning, lube-oil flushing and re-inspection. Machine restored to acceptable ISO vibration limits and returned to service.
- **HP gas compression system (120 MMSCFD, 5,500+ psi):** developed and drove the restoration strategy for the facility's most critical train, safeguarding continuity of gas export to the terminal.
- **10-year high filter clog rate on the lube-oil system:** identified the problem during a systematic system review, tracing a decade of high differential pressure, rapid filter clogging and element collapse to undocumented as-built versus as-designed discrepancies. Introduced an elemental **contamination tracker** with node-by-node sampling to isolate the ingress zones. Driving close-out through run-down tank flushing, reversion to the design filter elements and correctly sized control valves of the right specification, restoring the lube-oil system to design conditions.
- **Depropaniser pump chronic failures (BRT):** RCFA plus an **API 610** review of hydraulic selection, mechanical seal configuration and operating envelope proved the pump was running outside its design point; corrective actions substantially extended run-life and closed out a repeat-failure bad actor.
- **Foundation cracking across multiple pumps and generators (BRT):** led the RCA, established the causal chain and implemented structural reinforcement that restored integrity to the affected rotating equipment foundations.
- **Solar Taurus 60 gas turbine generator failure-to-crank:** supported diagnosis and corrective action with the OEM, restoring power generation availability to the terminal.
- **Firewater diesel engine vibration (UBIT):** resolved a long-standing vibration problem on a safety-critical package and validated performance against pump-curve criteria during deliverability testing.
- Work with operations, maintenance, inspection and electrical teams to resolve **machinery vibration problems across pumps, diesel engines and electric motor drivers** on several offshore platforms and onshore plants — field diagnosis through spectra, phase and alignment checks, corrective action, and post-repair verification against ISO limits.

Risk, Change Review, Projects & Start-up

- Contribute to **risk assessments, HAZID and safety reviews** for machinery interventions, and prepare Lessons Learned under a **PDRR (Prevent, Detect, Respond, Recover)** framework to prevent recurrence.
- Review equipment and system changes (MOC) for their effect on reliability, operability and maintainability; analyse deviations from specification and recommend accept/reject positions.
- Developed and now steward the facility **Pre-Startup Safety Review (PSSR)** process for turbines and compressors — verifying scope completion and safe restart readiness after maintenance and shutdowns.
- Support shutdown and turnaround planning with an emphasis on **optimising machinery scope and duration**, and provide start-up, commissioning and OEM coordination for critical rotating equipment.
- Provide discipline support to the **120 → 240 MMSCFD expansion**, reviewing machinery selection and integration of new equipment into the existing gas processing and NGL trains.

Manager, Business Innovations, Technology & Efficiency — Jul 2022 – May 2025

First Independent Power Limited (FIPL) / Sahara Power Group — multi-site gas turbine power generation fired on natural and associated gas

- Led **Reliability-Centred Maintenance (RCM)** deployment across the generating fleet, improving availability and MTBF of critical rotating equipment through revised strategies and defect elimination.
- Supervised **gas turbine performance testing to ASME PTC-22** — power output, heat rate and thermal efficiency verification — and used the results to drive degradation-recovery and washing decisions.
- Managed OEM **Factory Acceptance Tests (FAT)** for critical rotating equipment, witnessing performance and mechanical running tests and closing out punch items before shipment.
- Seconded as **Lead Mechanical/Reliability Engineer for the start-up of the Imo River Associated Gas Gathering (AGG) plant**, an upstream associated-gas facility — pre-commissioning checks, machinery alignment and verification, first fire and load-up support.
- Designed and deployed **IIoT-based condition monitoring and predictive maintenance** for gas turbines and compressors, enabling real-time machinery health visibility and delivering ~10% operational cost savings.
- Cut compressor wash downtime by **more than 24 hours** per event through procedure and scope optimisation.

Team Lead, Efficiency & Innovation — Sep 2019 – Jun 2022

First Independent Power Limited (FIPL) / Sahara Power Group

- Led major **gas turbine and compressor overhauls** — planning, clearance verification, rotor and bearing inspection, alignment and re-commissioning — delivering annual savings above NGN 10 million through improved spares management and maintenance planning.
- Chaired cross-functional **RCA investigations** into major turbomachinery failures and drove corrective actions that reduced repeat failures.

- Built real-time plant monitoring dashboards giving operations and maintenance immediate visibility of machinery health, vibration trends and process deviations.
- Coordinated emergency start-ups of critical turbine systems, minimising downtime and restoring generation.

Performance / QA-QC Engineer → Asst. Team Lead, Mechanical Maintenance — Nov 2015 – Sep 2019

First Independent Power Limited (FIPL) / Sahara Power Group

- Supervised maintenance of high-value turbomachinery and process equipment; delivered quality control during major inspections and overhauls to OEM and industry standards.
- Applied condition monitoring — **vibration analysis, lube-oil analysis, thermography, dissolved gas analysis and motor current signature analysis** — to detect anomalies and prevent unplanned downtime.
- Contributed to review and update of operations and maintenance procedures, strengthening safety protocols and process efficiency.

Graduate Engineer — Mechanical Maintenance — Aug 2014 – Aug 2015

Sahara Power Group — Egbin Power Plc (1,320 MW thermal station)

- Participated in overhauls of steam turbines, gas turbines, diesel engines and balance-of-plant equipment (pumps, compressors, valves, steam traps, accumulators); gained hands-on exposure to demineralisation, water treatment, desalination and lube-oil/water chemistry systems.

Junior Engineer (Intern) — Drillog Petro-Dynamics Ltd — Jun – Dec 2010

- Maintained directional drilling motors and auto-torque equipment and performed QA/QC inspection of mud motors prior to rig deployment — first exposure to oil & gas service operations.

Mechanical Trainee — Industrial Development Centre, Owerri — 2008 & 2009

- Machining, bench work, precision alignment of rotating components and fault diagnosis using basic root cause techniques.

EDUCATION

M.Eng Mechanical Engineering (Thermo-Fluids) — in view — **Rivers State University, Port Harcourt**

B.Eng Mechanical Engineering — Second Class Upper (2:1), 2012 — **Federal University of Technology, Owerri (FUTO)**

ROTATING EQUIPMENT PORTFOLIO — OEMS & MODELS SUPPORTED

Gas Turbines	Siemens SGT-600 • Solar Saturn, Centaur, Taurus and Mars families • GE Frame 5 (MS5001), Frame 9E (MS9001E) combined cycle and GT13E2 • Performance verification to ASME PTC-22; borescope, hot-section and mechanical maintenance support
Centrifugal Compressors	Siemens legacy Dresser-Rand centrifugal compressor family • HP trains to 5,500+ psi (120 MMSCFD) and LP trains to 1,440+ psi (130 MMSCFD) • Recompressors • Anti-surge and turbomachinery control (Honeywell CCC) • Seal-gas, dry-gas seal and lube-oil systems
Expanders & Cryogenic Pumps	Turbo-expander / booster-compressor trains • Cryogenic pumps — Cryostar and Ebara Cryodynamics (Elliott Ebara) • NGL fractionation: de-ethaniser, de-propaniser, de-butaniser • Sales gas (C1/C2) and NGL (C3+) export systems
Pumps, Drivers & Auxiliaries	API 610 centrifugal process pumps • Firewater diesel engine packages and deliverability testing • Steam turbines (Egbin 1,320 MW station) • Screw and reciprocating air compressors • Electric motor drivers and gearboxes • Lube-oil and seal-oil consoles, run-down tanks, coolers, filtration • Bearings, couplings, mechanical seals

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- ASQ Certified Reliability Engineer (CRE), 2024
- Vibration Analysis Category I — Istec Academy, 2024
- Reliability Engineering Principles — Beina M&R, 2024
- Gas Turbine Performance — Entropy-Tech, 2021
- Gas Turbine Mechanical Maintenance — Masaood John Brown, 2018
- Machinery Vibration Analysis & Alignment — O-Secul, 2019
- GE Frame 9E Combined Cycle — SEPCO-Pacific, 2015
- BOSIET with CA-EBS (2029) & Offshore Safety Permit (OSP)
- HSE Level I, II & III (OSHA) — HSETrain Intl., 2020
- QHSE, Risk & Sustainability Mgmt — IE Safetainability, 2024
- AI Advanced Applied Programme — Digital Regenesys, 2026
- IBM AI Engineering, 2023 & Generative AI and Agents — Johns Hopkins, 2026

PROFESSIONAL MEMBERSHIPS & RECOGNITION

- American Society for Quality (ASQ), 2024
- Society of Petroleum Engineers (SPE), 2022
- American Society of Mechanical Engineers (ASME), 2021
- COREN Registered Engineer, 2019
- Nigerian Society of Engineers (NSE), 2017
- Safety Advocate of the Year — FIPL, 2023
- Safety Sentinel Award — FIPL Trans-Amadi, 2023
- 1st Runner-Up — Sahara Group Innovation Hackathon, 2020
- Chairman's Recognition for Innovation — Sahara Group, 2019

PERSONAL DETAILS

- Date of birth: 25 September 1988
- Nationality: Nigerian
- Passport: B03319482 (Nigeria)

Availability: immediate mobilisation for international assignment. **References:** available on request.